

Sri Lanka Institute of Information Technology



IT3021 – Data Warehousing and Business Intelligence

Year 3, Semester 2 - 2025

Assignment 2

IT23258730

Step 1 – Data Source Description

The data warehouse used for this assignment is developed in Assignment 1. The warehouse follows a star schema design and consists of one fact table and three dimension tables.

Fact Table:

Fact_Transactions stores transactional data including amount and foreign keys referencing dimension tables.

Dimension Tables:

Dim_Customer – stores customer related information

Dim_Merchant – stores merchant related information

Dim_Date – stores date related information

The fact table is connected to all dimension tables using surrogate keys. This structure enables analytical queries and OLAP operations.

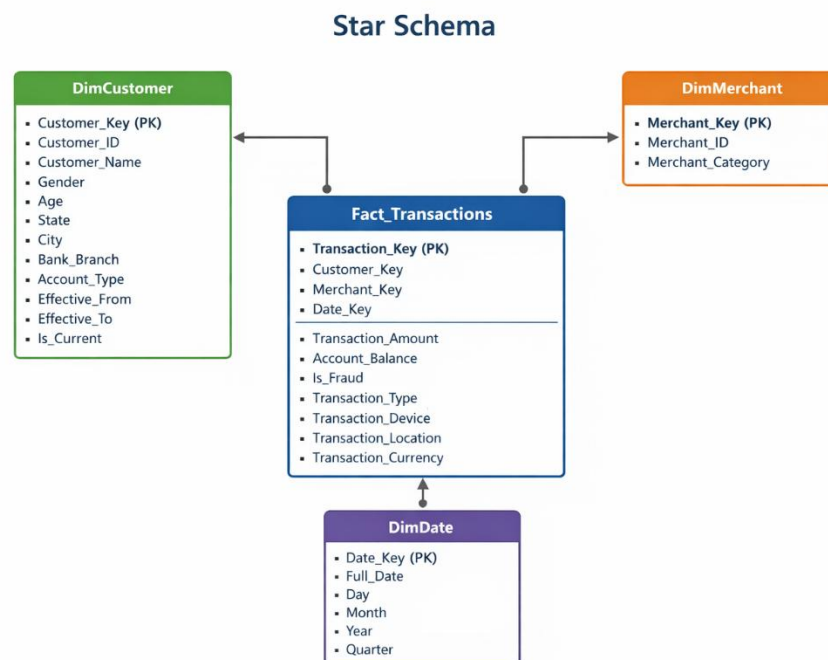
Relationships:

Fact_Transactions.Customer_Key → Dim_Customer.Customer_Key

Fact_Transactions.Merchant_Key → Dim_Merchant.Merchant_Key

Fact_Transactions.Date_Key → Dim_Date.Date_Key

The schema follows a star schema design where the fact table is at the center and dimension tables are connected around it.



Step 2: SSAS Cube implementation

The data cube was created using SQL Server Analysis Services (SSAS) to support analytical processing on the Bank Fraud Data Warehouse. The cube enables efficient querying and analysis of transaction data from multiple perspectives.

Data Source

A data source was created to establish a connection between the cube and the data warehouse database.

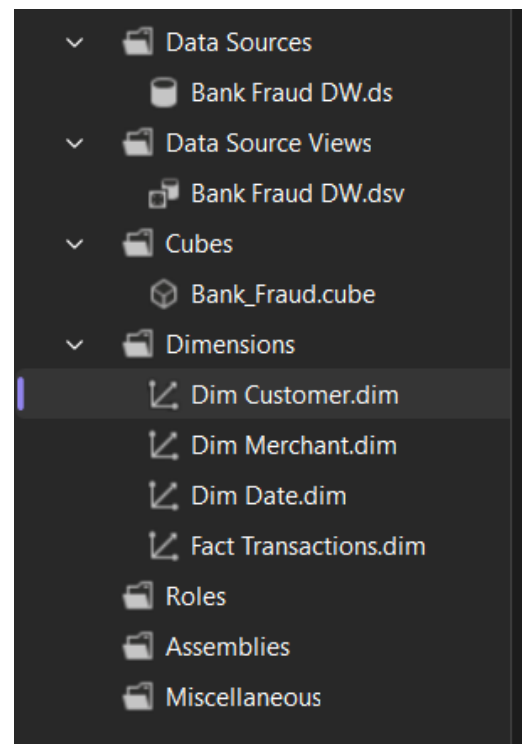
- Data Source Name: **Bank Fraud DW**
- Database: SQL Server Data Warehouse
- Purpose: To provide data for cube processing

Data Source View (DSV)

A Data Source View was created to represent the schema of the data warehouse.

The following tables were selected:

- Fact_Transactions (Fact Table)
- Dim_Customer (Dimension Table)
- Dim_Merchant (Dimension Table)
- Dim_Date (Dimension Table)



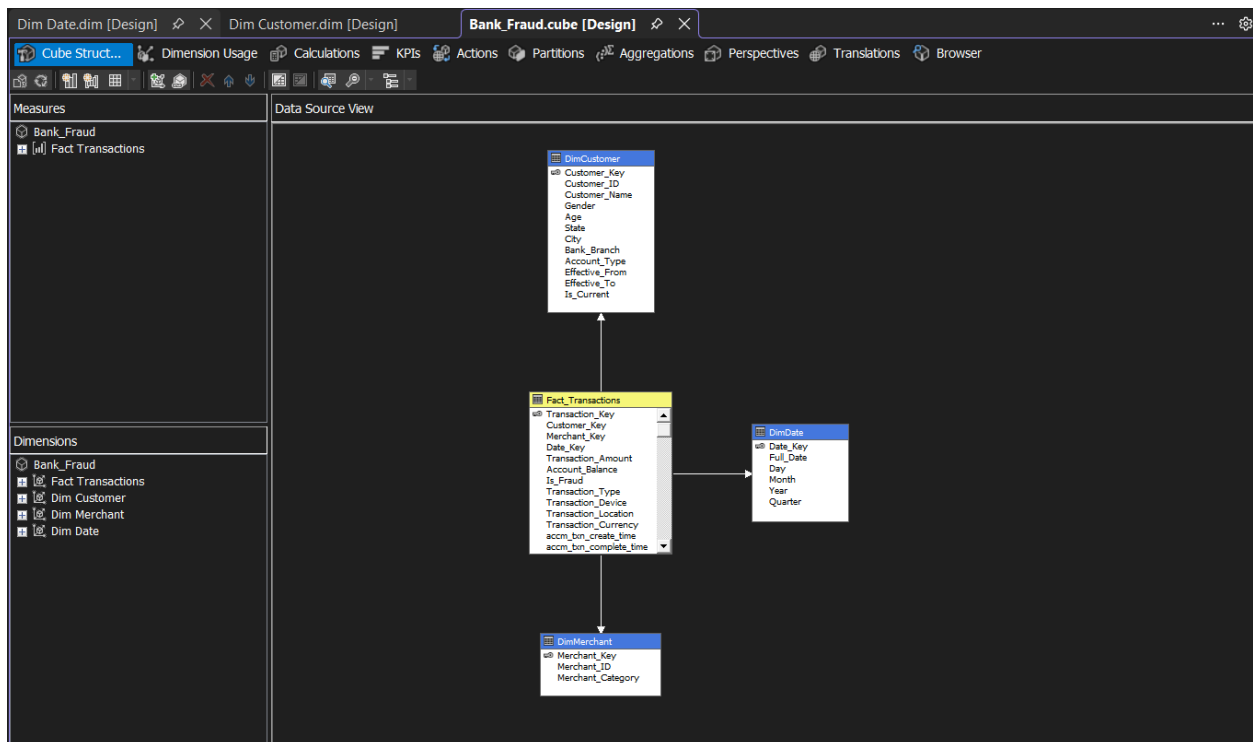
Cube Creation

A cube named **Sales Cube** was created using the existing tables in the Data Source View.

- Fact Table: Fact_Transactions
- Dimensions: Customer, Merchant, Date

The cube organizes transactional data for analytical queries.

This forms a **star schema**, where the fact table is connected to dimension tables using foreign keys.



Measures

Measures are numeric values used for analysis. The following measures were added:

- **Total Transaction Amount (SUM)**
 - Represents the total value of transactions
 - Fully additive measure
- **Total Transaction Account_Balance (SUM)**
 - Represents the total account balance

- **Transaction Count (COUNT)**
 - Represents the number of transactions
- **Average Processing Time**
 - Based on txn_process_time_hours

These measures support performance and fraud analysis.

Dimensions

Dimensions provide different perspectives for analyzing measures.

Customer Dimension

- Customer_Key
- Customer_Name
- City
- State
- Bank_Branch

Merchant Dimension

- Merchant_Key
- Merchant_Name
- Location

Date Dimension

- Date_Key
- Year
- Month
- Day

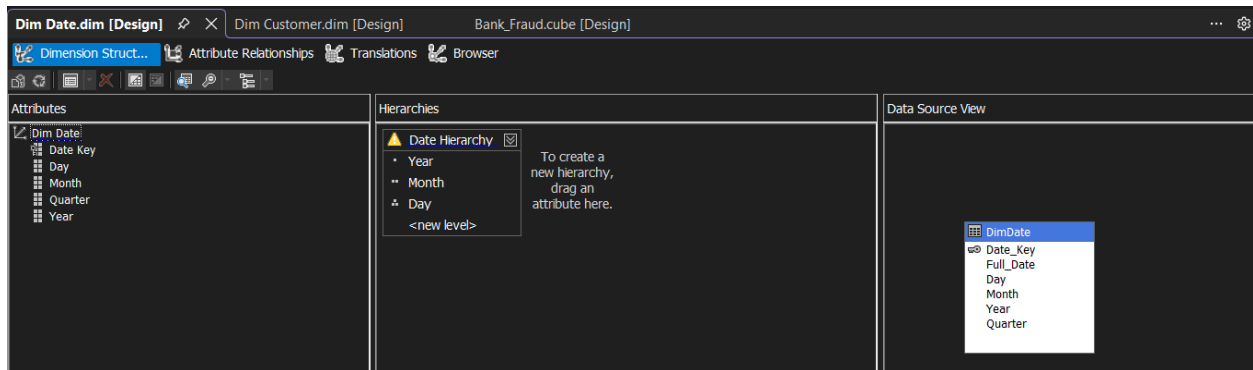
Hierarchies

Hierarchies were created to support drill-down analysis.

Date Hierarchy (Mandatory)

- Year → Month → Day

This hierarchy allows users to analyze data at different time levels.

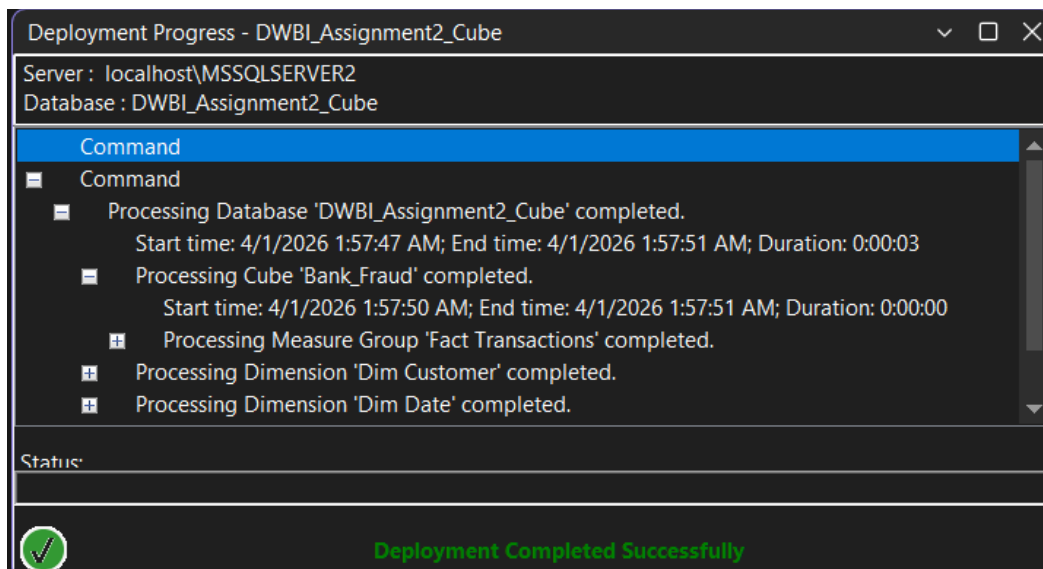


Customer Location Hierarchy

- State → City → Bank_Branch → Customer_Name

This hierarchy supports geographical analysis of transactions.

Deploy cube



Step 3: Demonstration of OLAP operations

Microsoft Excel was connected to the SSAS cube to perform OLAP (Online Analytical Processing) operations. Pivot Tables were used to analyze the multidimensional data stored in the cube and to demonstrate different OLAP functionalities.

Creating Pivot Tables

Pivot Tables were created using the connected cube data. Fields from dimensions and measures were dragged into Rows, Columns, Values, and Filters areas to perform different OLAP operations.

Demonstration of OLAP Operations

1. Roll-Up Operation

The roll-up operation aggregates data from a detailed level to a higher level.

Steps followed:

- Dragged **Year (Date Dimension)** into the Rows area
- Dragged **Total Transaction Amount** into the Values area

Result:

- Displayed total transaction amounts summarized at the Year level

Explanation:

This operation provides a high-level summary of data by aggregating transaction values over time.

The screenshot displays an Excel spreadsheet with a PivotTable and the PivotTable Fields task pane. The PivotTable, titled 'Total Transaction Amount by Year', is located in the range A1:B3. It shows the total transaction amounts for the years 2024 and 2025, with a grand total of 998483070.2. The PivotTable Fields task pane is open on the right side of the screen, showing the fields available for the report. The 'Dim Date.Year' field is selected for the Rows area, and the 'Transaction Amount' field is selected for the Values area. The 'Roll up' button is visible in the bottom left corner of the task pane.

Row Labels	Total Transaction Amount by Year
2024	499156141.9
2025	499326928.3
Grand Total	998483070.2

2. Drill-Down Operation

The drill-down operation allows users to navigate from higher-level data to more detailed levels.

Steps followed:

- Used the existing Year-level PivotTable
- Expanded the hierarchy using the “+” (expand) option
- Viewed data at **Month and Day levels**

Result:

- Displayed detailed transaction data by Month and Day

Explanation:

This operation enables detailed analysis by exploring lower levels of the hierarchy.

Row Labels	Sum of Transaction Amount
2024	499156141.9
1	39507154.3
10	1073164.88
11	2040596.56
12	1490612.44
13	1615823.58
14	1192000.12
15	1353277.82
16	1634028.98
17	1301057.78
18	1412704.12
19	1184205.1
2	250049.38
20	770748.4
21	1598127.38
22	1278127.36
23	1989697.14
24	1498920.16
25	1541256.06
26	1576758.18
27	1249230.64
28	2105900.8
3	1616596.06
4	1333003.68
5	891466.92
6	2461765.62

3. Slice Operation

The slice operation filters data based on a single dimension.

Steps followed:

- Added **Is_Fraud** to the Filters area
- Selected value **1 (Fraud transactions only)**

Result:

- Displayed only fraudulent transaction data

Explanation:

This operation extracts a subset of data based on one condition for focused analysis.

A	B	C	D	E	F	G
Fact Transactions.Is Fraud	True					
Row Labels	Sum of Fact Transactions Count	Sum of Transaction Amount				
2024	526	25013642.3				
2025	524	27483557.36				
Grand Total	1050	52497199.66				

PivotTable Fields

Choose fields to add to report:

Search

☒ Dim Date.Year

☒ Fact Transactions Count

☒ Fact Transactions.Is Fraud

☒ Transaction Amount

More Tables...

Drag fields between areas below:

Filters

Fact Transactions.Is Fra...

Columns

Σ Values

Rows

Dim Date.Year

Σ Values

Sum of Fact Transaction...

Sum of Transaction Am...

☐ Defer Layout Update

Update

< >

Roll up

Drill Down

Slice

Dice

Pivot

+

4. Dice Operation

The dice operation filters data using multiple conditions.

Steps followed:

- Added **Is_Fraud** as filter
- Added **State** as filter
- Rows: Merchant_Category
- Values: Total Transaction Amount

Result:

- Displayed transaction data for fraudulent transactions in appropriate state

Explanation:

This operation selects a smaller subset of data by applying multiple filters across dimensions.

A	B	C	D	E	F	G	H	I	J	K	L
Fact Transactions.Is Fraud	True										
Dim Customer.State	Haryana										
Row Labels	Sum of Fact Transactions Count	Sum of Transaction Amount									
Electronics	4	191263.88									
Entertainment	2	41955.18									
Groceries	4	117386.32									
Health	6	195405.58									
Restaurant	4	155014.72									
Grand Total	20	701025.68									

5. Pivot Operation

The pivot operation rotates the data to provide different analytical perspectives.

Steps followed:

- Initial view:
 - Rows → State
 - Columns → Merchant_Category
 - Values → Total Transaction Count
- Then rotated the data:
 - Rows → Merchant_Category
 - Columns → State

Result:

- Displayed the same data from a different orientation

Explanation:

This operation changes the layout of the data without modifying the underlying values, enabling flexible analysis.

A	B	C	D	E	F	G	H	I	J
Sum of Fact Transactions Count	Column Labels								
Row Labels	Clothing	Electronics	Entertainment	Groceries	Health	Restaurant	Grand Total		
Andaman and Nicobar Islands	108	102	124	102	86	100	622		
Andhra Pradesh	102	88	104	92	98	106	590		
Arunachal Pradesh	86	106	80	114	92	112	590		
Assam	92	90	68	122	104	100	576		
Bihar	96	106	92	74	124	80	572		
Chandigarh	98	92	84	90	114	90	568		
Chhattisgarh	100	106	82	106	96	102	592		
Dadra and Nagar Haveli and Daman and Diu	84	100	124	106	114	80	608		
Delhi	100	88	110	84	136	116	634		
Goa	78	100	94	76	66	114	528		
Gujarat	94	96	108	100	98	120	616		
Haryana	80	84	82	100	104	86	536		
Himachal Pradesh	76	120	104	92	98	90	580		
Jharkhand	94	96	94	106	94	120	604		
Karnataka	92	80	114	120	108	94	608		
Kerala	80	98	80	118	76	84	536		
Lakshadweep	98	106	102	90	66	100	562		
Madhya Pradesh	98	92	120	94	88	88	580		
Maharashtra	130	110	110	92	94	122	658		
Manipur	76	80	108	84	110	134	592		
Meghalaya	106	100	106	102	122	116	652		
Mizoram	104	82	64	108	108	112	578		
Nagaland	108	94	100	106	106	102	616		
Odisha	92	122	98	54	108	102	576		
Puducherry	78	110	92	114	124	94	612		
West Bengal	76	110	90	90	80	100	550		

PivotTable Fields

Choose fields to add to report:

Search

Dim Customer.State

Dim Merchant.Merchant Category

Fact Transactions Count

More Tables...

Drag fields between areas below:

Filters

Columns

Dim Merchant.Merchant...

Rows

Dim Customer.State

Values

Sum of Fact Transaction...

Defer Layout Update

Update

STEP 4 – Power BI Reports Documentation

Data Preparation

The dataset was first loaded into Microsoft Power BI Desktop using the *Get Data* option.

The following steps were performed:

- Verified all tables (Fact and Dimension tables) were correctly imported
- Checked for missing or null values and ensured data consistency
- Ensured correct data types were assigned (e.g., Date, Text, Number)
- Removed unnecessary columns to improve performance

Data Modeling

After loading the data, relationships were created in the *Model view*:

- Connected Fact_Transactions table with Dimension tables such as:
 - Dim_Date
 - Dim_Customer
 - Dim_Merchant

- Dim_Location
- Established relationships using primary and foreign keys
- Ensured a proper star schema structure

This helped in enabling filtering and interactions between visuals.

DAX Measures

To support analysis, the following DAX measures were created:

- Total Amount = SUM(Fact_Transactions[Transaction_Amount])
- Transaction Count = COUNT(Fact_Transactions[Transaction_Key])

These measures were used across all reports to display aggregated insights.

Report 1 – Matrix Report

Steps:

- Inserted a **Matrix visual**
- Assigned fields:
 - Rows → State
 - Columns → Transaction_Type
 - Values → Total Amount
- Renamed the page as *Report 1 – Matrix*

Visual Design:

- Used tabular layout for clear comparison
- Enabled grouping by rows and columns

State	Bill Payment	Credit	Debit	Transfer	Withdrawal	Total
Andaman and Nicobar Islands	5,550,688.60	5,698,685.32	5,656,672.10	5,516,334.16	7,509,859.16	29,932,239.34
Andhra Pradesh	5,680,873.52	5,883,689.12	4,347,801.52	6,002,797.28	5,941,077.04	27,856,238.48
Arunachal Pradesh	5,508,729.74	5,355,803.34	6,383,862.48	5,213,943.82	5,852,982.44	28,315,321.82
Assam	6,481,896.94	6,262,104.08	4,750,465.42	6,423,330.34	5,214,571.92	29,132,370.70
Bihar	5,390,097.56	4,505,094.66	6,744,183.68	6,066,607.56	4,718,517.88	27,424,501.34
Chandigarh	6,570,932.48	6,861,045.62	5,506,954.14	4,812,974.06	5,245,190.66	28,997,096.96
Chhattisgarh	6,730,893.14	6,127,236.72	4,653,220.78	5,208,142.44	5,434,190.68	28,153,683.76
Dadra and Nagar Haveli and Daman and Diu	6,337,908.86	7,366,913.38	5,593,958.74	6,037,182.78	5,151,547.42	30,487,511.18
Delhi	7,824,784.62	5,647,111.70	5,478,281.58	6,946,371.26	6,368,623.60	32,265,172.76
Goa	4,695,362.10	4,872,341.94	5,103,071.70	5,574,126.96	5,922,956.48	26,167,859.18
Gujarat	6,730,693.08	5,424,405.94	7,156,935.48	5,954,997.08	5,424,185.92	30,691,217.50
Haryana	5,564,816.54	5,187,226.88	4,832,494.88	4,822,018.58	4,640,377.94	25,046,934.82
Himachal Pradesh	6,015,989.50	6,331,665.68	4,567,750.74	7,276,426.20	5,505,579.22	29,697,411.34
Jharkhand	5,715,713.90	5,519,021.42	4,624,482.80	6,031,349.64	5,964,100.88	27,844,668.64
Karnataka	5,377,029.44	6,878,075.34	7,585,657.24	4,620,317.54	5,971,974.60	30,433,054.16
Kerala	5,374,749.88	5,487,195.78	4,774,940.08	4,676,348.68	6,654,984.40	26,968,218.82
Lakshadweep	5,925,112.48	6,209,840.96	3,608,056.22	6,588,034.08	5,105,648.28	27,436,692.02
Madhya Pradesh	6,189,981.38	5,728,239.50	6,253,437.46	5,839,420.62	6,296,449.42	30,307,528.38
Maharashtra	5,753,788.68	7,496,610.90	7,502,614.24	4,799,268.02	7,468,106.06	33,020,387.90
Manipur	4,905,181.46	5,867,757.64	6,686,140.26	5,060,824.32	5,884,007.86	30,403,911.54
Meghalaya	7,224,208.22	6,081,395.58	5,900,788.10	6,701,785.10	6,931,022.34	32,839,209.34
Mizoram	5,118,643.16	5,641,452.10	5,948,206.28	5,704,437.24	4,717,251.56	27,129,990.34
Nagaland	6,851,191.32	5,639,007.30	6,858,237.40	6,634,763.10	5,880,456.38	31,871,655.50
Odisha	6,486,906.76	5,128,044.12	6,526,123.26	6,776,607.72	4,866,248.98	29,783,930.84
Puducherry	5,528,281.96	7,122,895.74	5,832,450.32	4,829,019.64	6,752,502.54	30,065,150.20
Punjab	5,818,395.34	5,935,758.76	4,636,538.76	6,265,169.50	5,308,714.50	27,964,576.86
Rajasthan	5,229,257.30	5,315,078.10	5,240,176.30	7,276,795.86	6,009,497.18	29,070,804.74
Total	201,449,288.82	203,679,663.48	194,813,202.56	199,260,180.76	199,280,734.56	998,483,070.18

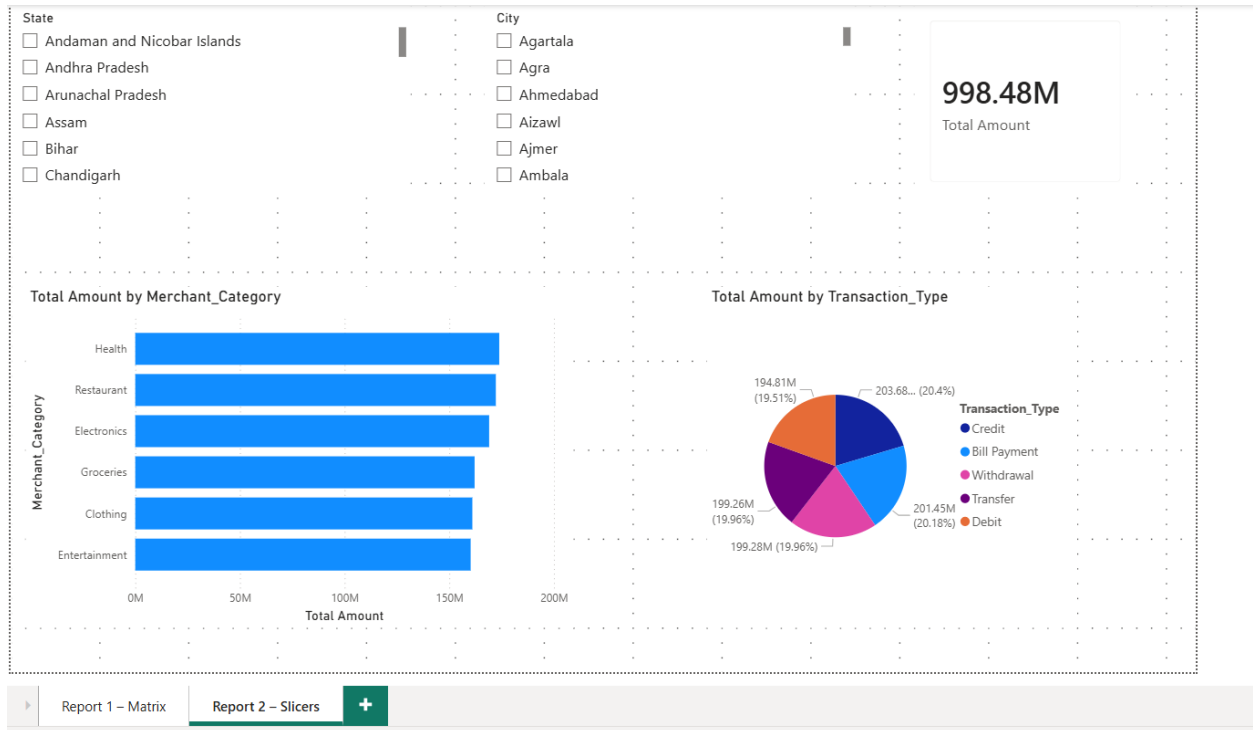
Report 2 – Slicers and Visuals

Steps:

- Added two slicers:
 - Slicer 1 → State
 - Slicer 2 → City
- Power BI automatically applied cascading filtering
- Added visuals:
 - Bar chart → Merchant vs Total Amount
 - Pie chart → Transaction_Type distribution
 - Card → Total Amount
- Renamed page as *Report 2 – Slicers*

Visual Design:

- Used multiple visuals for better insights
- Enabled interactive filtering using slicers



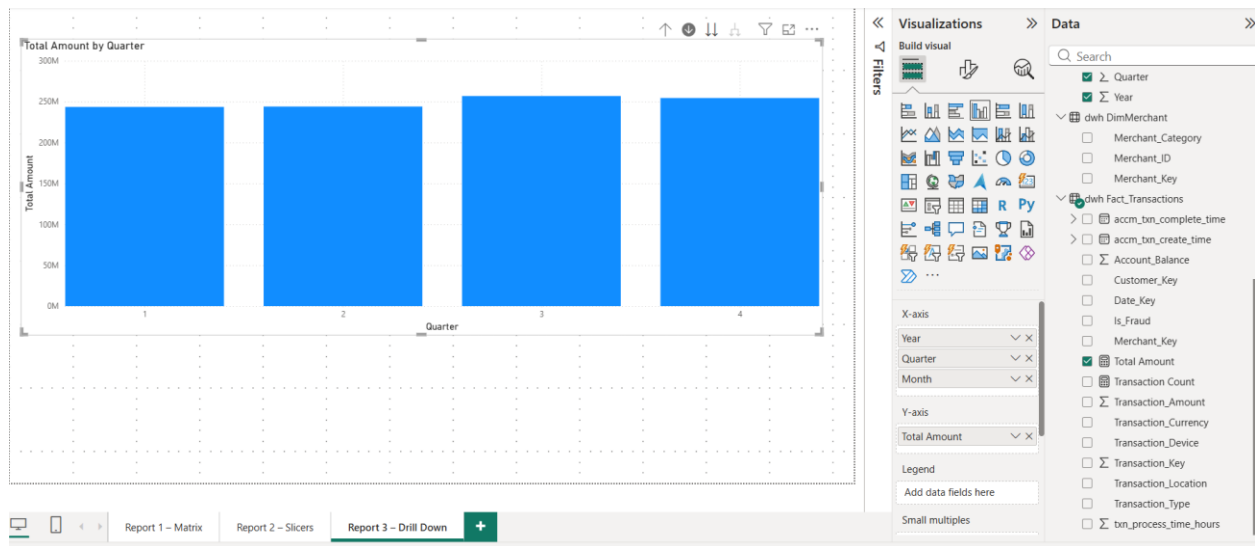
Report 3 – Drill-Down Report

Steps:

- Created hierarchy manually:
 - Added Year → Quarter → Month into Axis
- Inserted a Column Chart
- Assigned:
 - Axis → Year, Quarter, Month
 - Values → Total Amount
- Enabled Drill-down mode
- Renamed page as *Report 3 – Drill Down*

Visual Design:

- Used hierarchical navigation for time-based analysis
- Allowed users to drill from Year → Quarter → Month



Report 4 – Drill-Through Report

Steps:

- Created a new page named *Report 4 – Drill Through*
- Added State field to Drill-through filters
- Inserted visuals:
 - Table → Customer_Name, Total Amount
 - Chart → Merchant
- Added a Back button using:
 - Insert → Buttons → Back

Testing:

- Navigated from main report
- Right-clicked on a State value
- Selected Drill-through option
- Verified filtered data display

Visual Design:

- Focused on detailed view for selected state

- Enabled smooth navigation using back button

